

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 7

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1. Problem Title : Online Shopping Cart System

2. Problem Description :

Traditional shopping lacks convenience, proper order tracking, and easy product management.

The Online Shopping Cart System is designed to solve these issues by providing a digital platform where customers can browse products, add them to a cart, and place orders seamlessly. Sellers can manage their products, update prices, and monitor stock availability, while the system ensures smooth payment processing and order tracking.

This system improves user experience by making shopping faster, more organized, and accessible anytime, anywhere.

3. Identified Entities (Classes)

Class Name	Description
Customer	A person who uses the system to browse and purchase products
Product	An item that is available for sale in the system
Cart	A temporary storage where selected products are kept before purchase
Order	A confirmed purchase made by the customer
Payment	Handles the transaction process for orders
Seller	A user who adds and manages products in the system
Admin	A person who monitors and controls the overall system

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Customer	customerID, name, email, address	browseProducts(), addToCart(), placeOrder()
Product	productID, name, price, stock	getDetails(), updateStock()
Cart	cartID, items, totalPrice	addItem(), removeItem(), calculateTotal()
Order	orderID, orderDate, status	createOrder(), updateStatus()
Payment	paymentID, amount, paymentStatus	processPayment(), verifyPayment()
Seller	sellerID, productList	addProduct(), updateProduct(),manageStock()
Admin	adminID	manageUsers(), monitorOrders(), handleIssues()